BPH

Condensed rapid revision · facts already answered in Papers 1–4 point to that answer

highlighted number to memorise · **KEY POINT** and **ALGORITHM** boxes carry the reasoning · MCQ exam trivia · † added beyond the source notes · P2 topic = covered in full in that Paper's answer book

Chapter	Topic
Ch 148	Anatomy & Development of the Prostate
Ch 149	BPH: Aetiology, Pathophysiology & Natural History
Ch 150	Evaluation & Nonsurgical Management of BPH
Ch 151	Minimally Invasive & Endoscopic Management of BPH
Ch 152	Simple Prostatectomy
★	Putting it together: the BPH pathway

148 Anatomy and Development of the Prostate

1 · Development (embryology)

Why this matters: the androgen pathways that build the prostate also drive benign prostatic hyperplasia (BPH) and its drugs.

- **Origin**: urogenital sinus (UGS), endoderm in mesoderm. A female UGS forms prostate if given enough androgen. Incomplete cloacal division (imperforate anus, urorectal fistula) impairs development.
- **Dihydrotestosterone (DHT)**: †5α-reductase type 2 in UGS mesenchyme converts testosterone to DHT, the more potent androgen receptor (AR) ligand. **SRD5A2** loss leaves the prostate incomplete; castration or inactive AR leaves it absent. †The same enzyme is the target of 5α-reductase inhibitors (5-ARIs) in BPH.
- **Androgen is not rate-limiting**: testosterone is present at week 9 but budding starts at week 10. The gate is **AR demethylation** in the UGS (epigenetic checkpoint).

Week	Event
6–7	Cloaca divides into the ventral UGS and the dorsal anorectum
≈9	Testosterone activates **mesenchymal** AR, whose andromedins instruct the epithelium
10	Epithelial buds invade the mesenchyme; this fixes the adult ductal pattern

Mesenchyme instructs, epithelium responds

- **Lasnitzki–Mizuno** recombination: buds need **mesenchymal** AR; epithelial AR is not needed for initiation.
- **Cunha**: UGS mesenchyme turned adult bladder epithelium into prostate (**instructive** induction).
- Epithelial specification needs **Nkx3-1**, Edar and Wnt10b.
- **Andromedins** are androgen-induced stromal signals: **fibroblast growth factor 10 (FGF10)** (required), FGF7, Wif1 and Wnt2. FGF10 and retinoic acid have major roles.
- Branching needs **stromal** AR; Sonic hedgehog is the epithelial paracrine factor. Oestrogens are not needed for budding.

> **KEY POINT**
> McNeal: **re-awakening of embryonic induction** in ageing transition-zone stroma may initiate new glands, one theory of BPH.

- **Environmental**: gestational diethylstilboestrol (DES) causes squamous metaplasia; phthalates and dioxin (TCDD) cause agenesis or epimutations. TCDD-induced **hyperinnervation** gives hyperactive smooth muscle and frequency, perhaps explaining variable α-blocker response.

2 · Zonal anatomy (McNeal)

Why this matters: BPH and cancer arise in different zones.

- P1 Anatomy of prostate and neurovascular bundles: clinical and surgical concepts in localised prostate cancer — *zones (PZ 70% glands/70% cancers, CZ 25%, TZ 5%/BPH, AFMS), 35° urethral bend, surgical capsule, fasciae, NVB at 5/7 o'clock, urethral/capsular arteries, Santorini plexus, lymph nodes*

Not in the paper answer:

- Batson plexus route to spinal metastases; TZ gives ≈20% of cancers.
- The urethra bends **≈35°** at the verumontanum, where **13–15** ducts open dorsolaterally. This is the distal limit of transurethral resection (TURP) and enucleation.
- The transition zone (TZ) has the shortest ducts and gives †≈20% of cancers; the peripheral zone (PZ) has the longest ducts.

> **KEY POINT**
> Trap: the central zone (CZ), around the ejaculatory ducts, is not the TZ. Short proximal ducts **enlarge** (BPH); long distal ducts **transform** (cancer). The enlarging TZ compresses PZ into the **surgical (pseudo)capsule**, the enucleation plane *See Ch 152*.

3 · Fascia, vessels †

- †The capsule is incomplete anteriorly and apically. The neurovascular bundles (NVBs) lie outside it, so enucleation spares them.
- †Prostatic veins join the vertebral (**Batson**) plexus, a route to spinal metastases.

4 · Histology and cell types

- Acini have basal and luminal epithelium; stroma has fibroblasts and smooth muscle.

Feature	Luminal (secretory)	Basal
Keratins	8, 18	5, 14
Marker	Prostate-specific antigen (PSA), NKX3-1	KRT5/14
AR	High	Low
Secretes	PSA, prostatic acid phosphatase (PAP)	Almost none

- **Urethra and proximal ducts** carry lung-like epithelium: **club** (SCGB1A1), **hillock** (KRT13) and neuroendocrine cells (CHGA; mostly neural crest–derived). Handover to NKX3-1-positive epithelium is the cellular "transition zone".
- BPH stroma contains prominent T, B and dendritic cells.

5 · Secretions and clinical markers

Why this matters: semen tests, antibiotic choice and markers.

- Prostatic fluid is **37–44%** of semen and acidic (pH ≈6.3). Seminal vesicle fluid (the majority, alkaline) supplies semenogelins.
- Kallikreins liquefy semen within **20–30 min**; incomplete liquefaction at 60 min is nonliquefaction.
- **Ion trapping**: macrolides concentrate in acidic fluid; inflammation alkalinises the fluid.
- PSA, prostate-specific membrane antigen (PSMA) and PAP all occur outside the prostate ("prostate-specific" is a misnomer). The mouse has no PSA gene.

Product	Key facts
PSA (KLK3)	Serine protease (19q13.4); liquefies semen
PSMA	Folate hydrolase; rarely deleted in cancer, hence PSMA-PET and ^{177}Lu-PSMA
PAP	First tumour marker MCQ (Huggins and Hodges, 1941)
Zinc	Highest of any tissue; inhibits mitochondrial aconitase, so citrate is secreted

KEY POINT

Benign cells: high zinc blocks aconitase, so citrate is exported. Cancer: **zinc is lost** and citrate is oxidised, so citrate is low (†the basis of MR spectroscopy).

149 Benign Prostatic Hyperplasia: Aetiology, Pathophysiology, Epidemiology and Natural History

1 · Definitions and the four overlapping entities

Why this matters: histology, size, obstruction and symptoms differ, and treatment depends on which the man has.

- P2 Pathophysiology, classification and management of LUTS. — *BPH vs BPE vs BPO vs BOO, ICS symptom phases, static/dynamic (α1A) components, capsule/TUIP, bladder hypertrophy then decompensation, one-third keep storage LUTS after surgery*

Not in the paper answer:

- Non-prostatic causes include detrusor ageing, polyuria, sleep disorders and neurogenic or primary bladder disease. Impaired contractility mimics obstruction.
- "**Prostatism**" is obsolete; Paul Abrams coined "LUTS" MCQ. "Hypertrophy" is wrong because BPH raises cell **number**, not size.
- Storage symptoms bother more than voiding; post-micturition dribble is commonest. Age-matched women report similar LUTS.
- †BOOI = PdetQmax − 2Qmax: **>40 obstructed, 20–40 equivocal, <20 unobstructed**.

Term	How measured
Benign prostatic hyperplasia (BPH): a histological increase in epithelial and stromal cell number in the periurethral/transition zone (TZ)	Histology
Benign prostatic enlargement (BPE): enlargement caused by BPH	Digital rectal examination (DRE), transrectal ultrasound (TRUS), MRI
Bladder outlet obstruction (BOO): obstruction from any cause (BPH, stricture, cancer, bladder neck sclerosis)	Pressure–flow (best); flow rate (indirect)
Benign prostatic obstruction (BPO): BOO caused by BPE	Pressure–flow
Lower urinary tract symptoms (LUTS): storage, voiding and post-micturition; not prostate-specific	International Prostate Symptom Score (IPSS)

- The four circles overlap only **partially**. BOO may exist without LUTS, BPE or BPH (for example, a stricture).

> **KEY POINT**
> Viva: "Is every man with LUTS and a big prostate obstructed?" No. Size, symptoms, flow and obstruction are separate; obstruction is diagnosed only by pressure–flow.

2 · Epidemiology

Why this matters: most ageing men have histological BPH; the question is who has symptoms or obstruction.

- At autopsy there is **no BPH under 30**; prevalence is ≈ **50% in the 50s**† and **88% in the 80s**. It is **similar across all ethnic and geographic groups**.
- Clinical prevalence depends on the definition; none is globally accepted. Prevalence matters more than incidence.
- BPH is described on three axes: **symptoms** (IPSS), **enlargement** and **obstruction**, plus bother and quality of life (QoL). The proposed volume cut-off is **>30 mL**; a community

composite is IPSS >7, maximum flow rate (Qmax) <15 mL/s and volume >20 mL.

- P3 P1 IPSS score; Methods to estimate prostate size — *IPSS items, bands 0–7/8–19/20–35, QoL item, no incontinence/pain items, LURN; DRE underestimates, TRUS ellipsoid, MRI, CT*

Not in the paper answer:

- General QoL (SF-36) barely changes with LUTS, so disease-specific tools are needed. DRE underestimates size by 25% to ≥50%; MRI reads ≈10% larger than TRUS.
- Qmax falls almost linearly with age (Olmsted median 20.3 mL/s at 40–44 years versus 11.3 mL/s at 75–79). Probability of obstruction: Qmax <10 high, 10–15 intermediate, >15 low.
- Qmax depends on voided volume and varies day to day; a penile cuff test approximates pressure–flow.

ALGORITHM

1. Total prostate volume (TPV) ≈ **25 mL in the 30s → 35–45 mL in the 70s**.
2. Transition zone volume (TZV) ≈ 15 → 25 mL.

3 · Aetiology

Why this matters: each proposed cause explains a drug target or a risk factor.

KEY POINT

BPH is an **accumulation** disease: volume = proliferation minus death. Established BPH shows equal or reduced replication and raised bcl-2. Androgens act chiefly by **suppressing cell death**, so withdrawal shrinks the gland. Early BPH may involve rapid proliferation.

- New glands otherwise form only in the fetus, hence **embryonic reawakening** of stromal induction.

3.1 Androgens: permissive, not causal

- Prepubertal castrates and men with androgen defects **never develop BPH**; withdrawal gives only **partial** involution. The two established determinants are **age** and **functioning testes**.
- Serum testosterone falls with age, but **intraprostatic dihydrotestosterone (DHT) and androgen receptor (AR) stay high**. DHT is **maintained, not elevated**; old "raised DHT" reports used cadaveric controls MCQ.
- Serum testosterone does not predict volume or prostate-specific antigen (PSA); only body mass index (BMI) tracks low testosterone. A higher baseline serum DHT predicted 20-year BPH risk.
- The prostate keeps AR for life; penile AR is lost after puberty. Testosterone and DHT act via growth factors, not as direct mitogens. Androgens are **anti-inflammatory**. AR CAG-repeat and vitamin D receptor polymorphisms matter little.

ALGORITHM

1. Testosterone enters the stromal cell.
2. **Type 2 5α-reductase** (nuclear membrane, interstitial fibroblast) converts it to DHT.
3. DHT acts on the stroma (autocrine) or diffuses to the epithelium (paracrine).
4. The DHT–AR complex binds DNA and switches on growth factors.
5. Growth factors drive epithelial survival and growth.

- ≈ **90%** of prostatic androgen is DHT of testicular origin; ≈10% is adrenal. DHT has higher AR affinity and forms a more stable complex.

- Withdrawal switches off androgen-dependent genes (PSA) and activates death and inflammatory genes.

Feature	Type 1	Type 2
Site	Skin, liver	Prostate **stromal fibroblasts**, some basal cells
5α-reductase deficiency	Normal	**Mutated**
Finasteride / dutasteride	Not inhibited / inhibited	Inhibited / inhibited
Role	Possibly cancer	Development and BPH growth

- Finasteride shrinks the gland as much as dual inhibition (≈castration), so type 1 is unimportant in BPH.
- SRD5A2 lies on **chromosome 2p23** MCQ. The V89L variant lowers and A49T raises activity, but links to BPH are inconsistent. Promoter methylation occurs in up to 30% of BPH.

3.2 5α-reductase inhibitor (5-ARI) resistance

- 5-ARIs are the **only medical therapy that reduces volume**.
- Resistance is **not** failure to lower DHT but focal lack of luminal atrophy in nodules, or stromal hyperplasia. Inflammation-linked mechanisms: club-cell transition with high nuclear factor-κB (NF-κB), which induces AR variants, and SRD5A2 silencing, which shifts survival to oestrogenic pathways.
- Proposed definition: volume unchanged, no atrophy, and DHT decreased.
- On a 5-ARI or androgen deprivation therapy (ADT), luminal cells become club-like (lineage plasticity).

3.3 Oestrogens

- In dogs, androgen plus oestradiol produces BPH. Oestrogen induces AR, sensitises the gland and increases stroma and collagen; both hormones inhibit cell death.
- Oestrogen rises with age relative to testosterone; larger glands go with higher oestradiol; adipose tissue aromatises.
- **Oestrogen receptor α (ERα) is stromal** (proliferation, raises FGF-2 and FGF-7); **ERβ is epithelial** (antiproliferative, anti-inflammatory). An ERβ agonist failed in a randomised controlled trial (RCT) for LUTS.

3.4 Cell death; stromal–epithelial interaction

- Apoptosis is active (needs RNA and protein synthesis) without immune activation. After castration, transforming growth factor-β (TGF-β) is the likely mediator.
- Extracellular matrix (ECM) regulates epithelial phenotype: epithelium grown on collagen stays differentiated.
- Hypothesis: a **stromal defect removes a brake** on epithelial proliferation. Periurethral interstitial fibroblasts (carrying type 2 5α-reductase) increase in BPH and expand the TZ; **no drug targets them**.
- CXC chemokines and interleukin-8 (IL-8) turn fibroblasts into myofibroblasts, which lay down collagen; the fibrosis causes lower urinary tract dysfunction.

3.5 Growth factors

Androgen permits; growth factors signal and are the likelier primary defect. The first prostatic growth factor found was basic fibroblast growth factor (FGF-2) MCQ.

Factor	Source and target	Role in BPH
FGF-2	Stroma (autocrine)	Raised 2–3-fold; induced by TGF-β and hypoxia
FGF-7 (keratinocyte growth factor, KGF)	Made by stroma; receptor **only on epithelium**	Leading stromal mediator of hormonal control
FGF-10	Smooth muscle to epithelium	Raised by androgens
TGF-β1	Stroma	**Inhibits epithelium** but is **mitogenic for fibroblasts**; BPH stroma shows a blunted response
Insulin-like growth factors (IGF-1, -2)	—	Transgenic IGF-1 causes enlargement; serum IGF-1 is unrelated to volume
Others	Epidermal growth factor (EGF), vascular endothelial growth factor (VEGF), FGF-17	

- **Hypoxia hypothesis**: hypoxia induces hypoxia-inducible factor 1 (HIF-1), raising FGF-2 and FGF-7; nodules grow and become more hypoxic. Vascular disease (diabetes) may be a risk factor.
- α-blockade causes apoptosis in some models.

3.6 Inflammation

- **Activated T cells** secrete VEGF, heparin-binding EGF and FGF-2. IL-2, IL-7 and interferon-γ drive the stroma; senescent epithelium secretes IL-8; cyclo-oxygenase-2 (COX-2) is raised.
- Autoimmune disease increases BPH, and tumour necrosis factor (TNF) antagonists are associated with lower BPH incidence.
- **Inflammation correlates with enlargement and 5-ARI resistance**, but causality is unproven.

3.7 Metabolic syndrome, lifestyle

- Metabolic syndrome causes prostatic inflammation, leading to LUTS and BPH, via aromatisation, **hyperinsulinaemia driving sympathetic overactivity**, and low high-density lipoprotein (HDL).
- **Waist circumference** predicts prostatectomy and LUTS independently of BMI. BMI >35 versus <25 gives an odds ratio (OR) of **3.5** for a prostate ≥40 mL. Non-alcoholic fatty liver disease (NAFLD) is associated.

Factor	Direction
Diabetes, fasting glucose >110 mg/dL, obesity, central adiposity, hypertension (LUTS)	Increases risk
Physical activity; moderate alcohol	Lowers risk (alcohol lowers BPH, **not LUTS**)
Smoking, diet, vasectomy, sexual activity	No consistent effect (**vasectomy is not a risk factor**)
Jewish religion	More prostatectomy, not more BPH (access to care) MCQ

- **Depression** is strongly linked to LUTS; LUTS severity correlates with erectile and ejaculatory dysfunction.

- **Drugs that worsen LUTS**: α-sympathomimetic cold remedies; antidepressants, antihistamines and bronchodilators (+2–3 IPSS points). Anticholinergic and adrenergic drugs can precipitate retention.

3.8 Genetics and others

- In young men with large glands (>37 g), first-degree relatives have ≈ **4-fold** risk of surgery (brothers ≈6–8-fold). Inheritance is **autosomal dominant**; ≈ **50%** of men operated on before 60 have heritable disease, versus ≈9% after 60.
- **Familial BPH** shows early onset, a **large gland**, normal androgens and a normal 5-ARI response. Genetic risks of BPH and cancer correlate.
- Only **dogs and humans** get BPH; the seminal vesicle is never hyperplastic. Non-androgenic testicular factors may contribute.

4 · Pathology

Why this matters: nodule site and make-up explain lobe patterns and variable drug response.

- **Every nodule arises in the TZ or the periurethral region**. TZ: two glands outside the preprostatic sphincter, ducts at the urethral angle near the verumontanum. Periurethral zone: inside the sphincter, glands parallel to the urethra.
- The **middle lobe** is **periurethral**, because there is no TZ at the bladder neck MCQ. Patterns are middle, lateral, trilobar and posterior commissural (median bar).

KEY POINT

Only humans have a **capsule**, transmitting expansion onto the urethra; dogs get BPH but rarely obstruct. Transurethral incision of the prostate (TUIP) relieves obstruction without changing volume. Hence **size does not correlate with obstruction**.

Feature	Periurethral nodules	TZ nodules
Composition	Purely stromal (early), like **embryonic mesenchyme**	Glandular, tall epithelium
Origin	Interstitial fibroblasts	New duct branches forming a new ductal system

- **Two phases**: for ≈20 years nodule number rises; then (glandular) nodules grow.
- Small resected glands are mainly stroma; large enucleated glands are mainly epithelial nodules. A variable stromal:epithelial ratio explains variable drug response.
- In symptomatic BPH, smooth muscle and connective tissue exceed epithelium. α-blockers target smooth muscle; 5-ARIs target epithelium.

5 · Pathophysiology of obstruction

Why this matters: many symptoms come from the bladder, so some persist after surgery.

ALGORITHM

1. Hyperplasia raises urethral resistance (static and dynamic components).
2. The detrusor compensates, at a cost to storage function.
3. Storage LUTS appear, with or without impaired emptying.
4. In a minority, the detrusor decompensates.

- The static component (bulk under the capsule, elastic ECM) is **not** reduced by α-blockade. The dynamic component is smooth muscle tone via **α1a** receptors (the most abundant subtype).
- BPH smooth muscle is **dedifferentiated**, with less smooth-muscle myosin heavy chain. The fall in stromal cells after ablation may take ≥1 year.
- Other regulators: endothelin, kallikrein–kinin and **phosphodiesterases 4 and 5 (PDE4/5)**, so PDE5 inhibitors help LUTS. Autonomic overactivity correlates with symptoms.
- Many LUTS arise from **obstruction-induced bladder change**. ≈ **One-third** of men keep voiding dysfunction (mostly storage) after relief.
- Storage changes: detrusor overactivity (DO) and lower compliance. Emptying changes: reduced contractility, and detrusor failure in a minority.

ALGORITHM

Three-stage remodelling:

1. Smooth muscle hypertrophy.
2. Compensation.
3. Decompensation (underactive bladder).

- Stimuli are stretch, pressure and **hypoxia/ischaemia**. The detrusor switches to a **secretory phenotype** that makes ECM, with mitochondrial and calcium-signalling dysfunction.
- **Trabeculation** reflects increased detrusor **collagen**; when severe, residual urine is significant. There is no accepted grading. Sequence: trabeculation, cellules, diverticula, failure. Ageing alone reproduces some changes.

KEY POINT

Storage LUTS persist after transurethral resection of the prostate (TURP) because the collagen-rich, overactive detrusor does not reverse, so counsel men with DO before surgery. Acute urinary retention (AUR) is **not** the inevitable end: many men in AUR have an adequate detrusor and a precipitant.

6 · Natural history

Why this matters: who progresses decides watchful waiting versus treatment.

- Longitudinal population studies give the best data (no selection).
- **Regression to the mean**: entry thresholds select men at a temporary extreme, so they seem to improve; stricter thresholds give larger effects. It is part of every placebo effect.
- **Restriction of range**: in trials, IPSS, Qmax and volume barely correlate; only TPV with TZV, and PSA with volume, still correlate.
- Outcomes: **direct** (symptoms, AUR, surgery) or **proxy** (Qmax, residual, volume).
- ≈ **40%** improve without treatment (watchful waiting behaves like placebo); active drug gives about twice the placebo improvement in each stratum.
- With mild LUTS ≈1 in 3 progress over 4 years; in the community regression equals progression.
- **Olmsted County**: IPSS rises ≈ **0.2–0.34 points/year** (steepest in the 60s); Qmax falls ≈ **2.1%/year** (faster if age ≥70 or Qmax <10); volume grows ≈ **0.6 mL/year** (≈1.9%/year, faster with higher PSA or volume). Volume >50 mL gives ≈3.5-fold odds of moderate–severe LUTS.

- PSA and volume are log-linear but imprecise individually. A perceptible improvement is ≈ −2 points at low baseline IPSS versus −6 at high baseline.

Predictor of progression	Predicts
Age	Everything; AUR rises linearly from age 40 to 80+
PSA ≥1.4–1.5 ng/mL	Growth, symptoms and flow, AUR, surgery
Volume >30–40 mL	AUR (≈3-fold), surgery, new LUTS
IPSS ≥8 or high bother	AUR, surgery
Qmax <10–12 mL/s	AUR, flow decline, treatment
Raised post-void residual (PVR)	Treatment failure, AUR

- **MTOPS** (Medical Therapy of Prostatic Symptoms) placebo arm: progression **4.5 per 100 patient-years** (17% at 4 yr), mostly an **IPSS rise ≥4** (14%); AUR 2%; invasive therapy 5%; **no renal insufficiency**.
- **PLESS** (Proscar Long-term Efficacy and Safety Study) placebo arm: symptoms drifted back to baseline, fastest in the top PSA tertile (3.3–10). Surgery was 10% at 4 yr, rising from ≈6% to ≈15% across PSA and volume tertiles.
- **REDUCE** placebo arm: volume 40–80 mL gave a hazard ratio (HR) of 1.67 for new LUTS; Qmax <10 also gave 1.67.

> **KEY POINT**
> PSA is a cheap surrogate for volume, the main driver of AUR and surgery; models with >100 predictors barely beat PSA alone.

7 · Complications

Why this matters: these are the absolute indications for surgery.

> **KEY POINT**
> Death, renal failure, stones and infection are now rare. The common events are **AUR and surgery**, predicted by age, symptoms, low flow and a large gland (or high PSA).

7.1 Acute urinary retention

- AUR was historically 25–33% of TURP indications. Precipitants: infection, overdistension, fluid, alcohol, sex, bed rest; prostatic infarction is **not** proven.

	Spontaneous AUR	Precipitated AUR
Trigger	None	Surgery, anaesthesia, catheterisation, sympathomimetics, anticholinergics, antihistamines
Later spontaneous AUR	15%	9%
Later surgery	**75%**	**26%**

- AUR incidence is ≈ **4.5–6.8/1000 person-years** in the community, 9.6 in Veterans Affairs (VA) watchful waiting, 18 in PLESS placebo and 6 in MTOPS. The hazard is constant over time.
- **Age and symptoms multiply risk**: in Olmsted, men in their 70s with IPSS >7 had 34.7/1000 person-years. Relative risks: IPSS ≥8 3.2; Qmax <12 3.9; volume >30 mL 3.0; age ≥70 up to

≈15.

- PSA >1.4 raises risk several-fold and volume >40 mL ≈3-fold. Best IPSS items: incomplete emptying, frequency, weak stream. Recurrence: older age, diabetes, neurological disease.

7.2 Other complications

- **Renal**: older series: ≈1 in 7 men at TURP had renal failure, with up to 6-fold complications and mortality. "**Silent prostatism**" is asymptomatic renal failure from BOO. MTOPS had none in >3000 men. †Catheterise first, watch for post-obstructive diuresis, let creatinine stabilise, then operate.
- **Bladder stones** are ≈ **8×** commoner because of stasis; renal stones are not.
- **Haematuria** reflects increased microvessel density; **finasteride** is first-line because it lowers VEGF. †Exclude cancer and stones first.
- **Urinary tract infection (UTI)**: a link with residual urine is unproven; MTOPS rate was 0.1/100 patient-years. **Incontinence** is overflow or urge (DO in up to half).
- **Mortality** has collapsed since the 1950s; storage LUTS are linked to higher all-cause mortality. Harm from delay is only indirect (smaller benefit from delayed TURP in the VA study).
- **Surgery** is a soft end point (provider, era); AUR is hard. In Olmsted, moderate–severe symptoms, Qmax <12, volume >30 and PSA ≥1.4 each gave ≈4× treatment; ≈1 in 4 men were treated by the eighth decade. In Baltimore, 0 risk factors gave 3% surgery and 3 gave 37%.

8 · Landmark trials

Why this matters: these studies supply most natural-history numbers.

Trial	Design	Result	Take-home
VA Cooperative Study	TURP vs watchful waiting, moderate symptoms	Failure RR 0.48; 5-yr 10% vs 21%; ≈36% crossed over; delayed TURP smaller gains	Watchful waiting safe; delay may reduce benefit
PLESS	Finasteride vs placebo, 4 yr	Placebo returns to baseline; AUR/surgery rise with PSA/volume	PSA/volume predict progression
MTOPS	Placebo vs doxazosin vs finasteride vs combination	Placebo progression 4.5/100 patient-years, mostly IPSS ≥4; no renal failure	Defined "clinical progression"
Olmsted County Study	Population cohort, 40–79	IPSS +0.2–0.34/yr, Qmax −2.1%/yr, volume +0.6 mL/yr; AUR 6.8/1000 person-years	Best natural history; age, IPSS, Qmax, volume predict AUR

150 Evaluation and Nonsurgical Management of Benign Prostatic Hyperplasia

1 · Definitions and the diagnostic problem

Why this matters: not every man with urinary symptoms is obstructed; find the cause first.

- Benign prostatic hyperplasia (BPH) is **histological** smooth muscle and epithelial proliferation in the **transition zone (TZ)**. It can lead to benign prostatic enlargement (BPE), then benign prostatic obstruction (BPO, a form of bladder outlet obstruction, BOO), then lower urinary tract symptoms (LUTS).
- LUTS are multifactorial: detrusor overactivity (DO) or underactivity (DUA), stricture, urinary tract infection (UTI), metabolic, cardiovascular or neurological disease.

> **KEY POINT**
> Bothersome LUTS are not BPH-related BOO until assessed. Aims: **grade severity and bother** and **identify the cause**. Symptoms, flow, post-void residual (PVR), volume and cystoscopy cannot separate obstruction from bladder dysfunction; only the **pressure–flow study** can (gold standard, reserved for when it changes surgery).

Guidelines: American Urological Association (AUA) BPH (2021); European Association of Urology (EAU) non-neurogenic male LUTS (2022).

2 · History and symptom assessment

Why this matters: history finds other causes; the score measures bother.

- **Comorbidities**: diabetes mellitus or insipidus, renal disease, heart failure, sleep apnoea, neurological disease (Parkinson, multiple sclerosis, stroke, spinal cord injury, disc disease), haematuria and prior lower-tract surgery.
- **Drugs**: loop diuretics, β-blockers, excess fluid and caffeine worsen storage; anticholinergics impair contractility.
- Assess **sexual function**. LUTS do **not** raise prostate cancer (PCa) risk (detection bias; Prostate Cancer Prevention Trial, PCPT, placebo arm).
- Voiding: weak stream, hesitancy, straining, intermittency, incomplete emptying, dribbling. Storage: frequency, urgency, urge incontinence and nocturia (**overactive bladder, OAB**).
- P3 IPSS score — *AUASI 7 items + QoL (0–6, WHO 1993), subscores not validated, does not diagnose BOO, VPSS/BII/OAB-q alternatives*

Not in the paper answer:

- The minimum perceptible change is **3 points**, but with a severe baseline **percentage** reduction is more reliable.
- The AUA Symptom Index (AUASI) and International Prostate Symptom Score (IPSS) total **0–35**: mild **0–7**, moderate **8–19**, severe **20–35**.

3 · Examination and DRE

Why this matters: examination detects retention and gives a first size estimate.

- Look for a suprapubic bladder (retention); penile meatal stenosis, phimosis, urethral mass or discharge; small testes (low testosterone, metabolic syndrome). Check **anal tone** and

perineal and lower-limb neurology.

- **Digital rectal examination (DRE)** estimates volume and screens for PCa (low sensitivity; with PSA it reduced overdiagnosis in ERSPC). It **underestimates prostate volume (PV) above 30 mL**. Positive predictive value is good at 30 mL, near-perfect at 50 mL; a flat gland suggests <30 mL.
- PV does not correlate with symptoms or obstruction, but the **30 mL** and **50 mL** cut-offs predict treatment response and progression.

4 · Frequency–volume charts and nocturia

Why this matters: a chart separates polyuria from a small bladder.

- International Continence Society (ICS) terms: a micturition chart records times; a frequency–volume chart (FVC) adds volumes over 24 h; a bladder diary adds intake, incontinence, pads and urgency.
- **Indication**: predominant **storage LUTS or nocturia**; it beats recall. **Duration**: **3 days** (adherence falls by 7). Charting has a training effect; keep habits routine.
- **Nocturia** is waking ≥1 time to void. **Nocturnal polyuria** is night output **>20%** of 24-h output (young) or **>33%** (over 65).

ALGORITHM

Nocturia

1. Complete an FVC.
2. If †global polyuria (>40 mL/kg/24 h), look for diabetes or excess intake.
3. If nocturnal polyuria, look for heart failure, sleep apnoea, evening fluids or diuretic timing.
4. If reduced capacity, think of OAB or BOO.
5. Treat the cause. If nocturia persists, consider **sleep apnoea** (AUA).

Nocturnal polyuria: no fluid 2 h before bed, no evening alcohol or caffeine, retimed diuretics, with or without **desmopressin**. †It risks **hyponatraemia**, so check sodium at baseline, 1 week and 1 month.

5 · Laboratory tests

Why this matters: urinalysis is mandatory; PSA also predicts progression.

- **Urinalysis is mandatory** for UTI, microhaematuria, glycosuria and proteinuria. Pyuria and nitrite are insensitive in painless LUTS (reference 10^5 colony-forming units/mL). Gross haematuria: exclude bladder cancer. **Cytology**: severe storage symptoms with dysuria, or smokers (carcinoma in situ).
- **Prostate-specific antigen (PSA)** gives PCa risk, a volume surrogate and progression risk. Test only if life expectancy is **>10 years** and it changes management.
- PSA predicting **PV >40 mL**: **>1.6** (50s), **>2.0** (60s), **>2.3 ng/mL** (70s); area under the curve ≈0.76–0.78. Baseline PSA predicts acute urinary retention (AUR), surgery, and change in symptoms and maximum flow rate (Qmax).
- **5α-reductase inhibitors (5-ARIs) lower PSA by 40–50% by 12 months**, so take a baseline first. †Double the PSA after ≥6 months; any rise from the nadir needs evaluation.
- **Creatinine and estimated glomerular filtration rate (eGFR)**: EAU if hydronephrosis, renal insufficiency or retention is suspected; AUA not explicitly. Renal failure was <1% over 4

yr (MTOPS); it raises surgical complications.

6 · Instrumental investigations

Why this matters: only pressure–flow proves obstruction.

PVR

- Measure by ultrasound soon after voiding (catheter: gold standard, invasive). No standard normal exists.
- A PVR **<30 mL** is insignificant; persistently **>50 mL** is important; **200–300 mL** may mean DUA and a poorer response; **>300 mL** is significant chronic retention.
- High PVR means BOO **or DUA**; it predicts symptom deterioration (MTOPS >39 mL; ALTESS >93 mL), **not surgery**. Rising PVR predicted AUR; ≥500 mL predicted recurrence.

Uroflowmetry

- Record **Qmax**, voided volume and pattern (average flow is less accurate). Repeat if voided volume is **<150 mL**; void in the preferred position.
- It **cannot distinguish BOO from DUA** (DUA in 9–48%). Qmax **10 mL/s** is more specific; **15 mL/s** is more sensitive.
- Optional at first; **recommended before active treatment** and for monitoring.

Urodynamics

- P2 Role of UDS before prostate surgery — *cystometry/PFS, BOOI and BCI, BOO = high pressure + low flow vs DUA, EAU indications, UPSTREAM: not routine*

Not in the paper answer:

- DO is common in BPH; cystometry identifies antimuscarinic responders. DUA predicts long-term surgical failure; contractility is independent of treatment and obstruction duration.
- Pressure–flow study (PFS) indications (EAU): failed invasive treatment; cannot void **>150 mL**; PVR **>300 mL**; age **>80** or **<50 years** with voiding LUTS; neurological disease or pelvic surgery. AUA: before surgery if **Qmax >10 mL/s**.

Imaging and endoscopy

- P1 Methods to estimate prostate size — *TAUS/TRUS ellipsoid, planimetry, MRI, CT; IPP grading*

Not in the paper answer:

- **Transrectal ultrasound (TRUS)** is the method of choice for PV. It is needed before a 5-ARI and **any BPH surgery**; a median lobe guides the choice of minimally invasive surgical therapy (MIST). PV predicts progression and AUR (ALTESS >49 mL), but BOO only weakly.
- **Intravesical prostatic protrusion (IPP) ≥10 mm** is high grade; it predicts need for therapy and a **failed trial without catheter (TWOC)**. Bladder wall thickness (BWT, 2 mm), detrusor wall thickness (DWT), ultrasound-estimated bladder weight (UEBW), penile cuff test and near-infrared spectroscopy (NIRS) are accurate, but **none replaces PFS**.
- Upper-tract ultrasound only for raised creatinine, large PVR, haematuria, UTI or stones; intravenous urography is obsolete.
- **Cystoscopy** neither diagnoses BOO nor decides treatment. Indications: gross haematuria or prior bladder cancer, recurrent UTI, stricture, prior surgery, and to **exclude a median lobe before prostatic urethral lift (PUL)**; in a young man with a small prostate, **bladder-neck contracture**. Voiding cystourethrogram (VCUG): stricture, diverticulum, reflux.

7 · Evaluation algorithms

Why this matters: they set the minimum work-up.

ALGORITHM

EAU (men ≥40 yr)

1. History (including sexual), symptom score, examination with DRE, urinalysis, PVR, and PSA if life expectancy >10 yr.
2. If DRE, PSA or urinalysis is abnormal, or neurological disease, follow the relevant guideline.
3. If not bothersome, no further work-up.
4. If PVR is significant, renal ultrasound with or without renal function.
5. If bothersome, FVC (storage or nocturia), then prostate ultrasound and uroflow if treating, then the medical algorithm.
6. Before surgery, endoscopy if it alters the choice; PFS for specific indications.

ALGORITHM

AUA

1. History, examination, IPSS, urinalysis, with or without PVR and flow.
2. If **PVR >300 mL**, manage as non-neurogenic chronic retention.
3. Counsel: behavioural, medical, procedural.
4. Reassess at **4–12 weeks**. If improved, follow up. If nocturia predominates, FVC and a drug. If incomplete, FVC, PVR and flow, urodynamics and cystoscopy, then the OAB pathway, surgery, or both.

8 · Watchful waiting (WW)

Why this matters: many men need only advice and review.

- **Indication**: mild–moderate uncomplicated LUTS, minimal bother. **Excluded**: gross haematuria, UTI, raised PSA or PVR, and neurological disease.
- WW is reassurance, education, lifestyle and monitoring; ≈half stay stable over 15 years. In the Veterans Affairs (VA) trial most WW men did not fail; ≈a quarter crossed over (high **bother**). Self-management sessions reduced failure. Review **yearly**.
- Lifestyle: fluids **1.5–2 L/day**, none 2 h before bed; less caffeine and evening alcohol; retime diuretics; **double voiding** and **urethral milking**; avoid constipation; bladder retraining (suppress urge, target ≥3 h or 200–400 mL).

KEY POINT

Treatment is driven by **bother and complicating features**, not by score or size alone.

9 · Medical therapy: choice

Why this matters: choice follows symptoms, size and risk.

- Drugs are first line without an **absolute surgical indication** (refractory or recurrent AUR, recurrent UTI, renal insufficiency, stones, recurrent haematuria *See Ch 151*).
- P2 BPH: medical management and surgical modalities — *drug classes and doses, MTOPS/CombAT/SMART, IFIS, ALFAUR, desmopressin, surgical options by size*

Not in the paper answer:

- Antimuscarinics need PVR <200 mL; avoid oxybutynin in the elderly; hexane-extract Serenoa is an EAU option.

ALGORITHM

EAU

1. Not bothersome: WW.
2. Nocturnal polyuria: lifestyle with or without desmopressin.
3. Storage predominant: antimuscarinic or β3-agonist.
4. If **PV >40 mL and long-term treatment** intended: 5-ARI with or without an α1-blocker or phosphodiesterase-5 inhibitor (PDE5I).
5. Otherwise α1-blocker or PDE5I; if storage symptoms remain, add an antimuscarinic or β3-agonist.

AUA: α-blocker (PDE5I if erectile dysfunction, ED) → **prostate >30 mL** → add 5-ARI → PDE5I trial → surgery.

KEY POINT

Reading trials: placebo improves scores and flow (a 4-week run-in is ideal); judge by a 3-point or percentage threshold, and Qmax as %. Goals: relieve symptoms, reduce BOO, improve emptying and **prevent progression**.

10 · α1-Blockers

Why this matters: usual first drug; pick it by side effects.

- P2 Tamsulosin; Silodosin. — *α1a ≈70% of prostatic α1, subtype table, dynamic component, onset, ejaculatory dysfunction, IFIS, silodosin renal dosing, greatest BOOI fall*

Not in the paper answer:

- **No α1-blocker is clearly more effective**; they differ in adverse effects (AEs). Doxazosin cut progression 39% (MTOPS), **not AUR or surgery**.
- **ALLHAT**: the doxazosin arm stopped for heart failure, so never use it as sole antihypertensive; **never use two α-blockers**.
- **Elderly**: dizziness causes falls and fractures; severe hypotension is mostly in the **first 8 weeks**, so check blood pressure in week 1.
- They do **not** shrink the gland. α1d blockade may increase capacity, reduce DO and improve bladder perfusion. α1b blockade causes vascular AEs; on the detrusor α1d is twice α1a.

Class	Drug	Dose	Notes
Non-selective	Phenoxybenzamine	10 mg bd	α1+α2; abandoned
Short-acting	Prazosin, indoramin	bd	Not used for LUTS
Long-acting	Terazosin	5–10 mg od	Start **1 mg at bedtime**, titrate
	Doxazosin immediate release (IR)	2–8 mg od	Start 2 mg at bedtime, titrate
	Doxazosin GITS (sustained release)	4–8 mg od	Start **4 mg**, **no titration**
Uroselective	Alfuzosin extended release (ER)	10 mg od	Clinically uroselective
	Tamsulosin	0.4 mg od (to 0.8)	α1a/α1d
	Silodosin	8 mg od	Most α1a-selective
	Naftopidil	25–75 mg/day	Only α1d-selective; Asia

KEY POINT

Titration: α1b blockade causes **first-dose hypotension** and syncope; a bedtime start and titration avoid it. Uroselective drugs are fixed-dose; doxazosin GITS starts at 4 mg without titration.

- Level 1a evidence: ≈double the placebo symptom gain, modest Qmax rise, **durable** benefit (PREDICT, MTOPS). Tamsulosin raises flow in hours and improves symptoms by 1 week.
- Tamsulosin plateaus at **0.4 mg** (0.8 adds only AEs); alfuzosin 15 mg equals 10 mg; terazosin works best at 10 mg. Naftopidil gives more storage relief.
- **AEs**: dizziness, asthenia, orthostatic hypotension (standing systolic fall ≥20 mm Hg with symptoms) and anejaculation. Blood pressure falls significantly only in uncontrolled hypertensives.
- **Sexual**: ED equals placebo. Ejaculatory dysfunction may be retrograde (bladder neck), **anejaculation** (emission failure) or central. Silodosin (odds ratio ≈32) > tamsulosin (≈9) > alfuzosin; younger men are more affected.

KEY POINT

Intraoperative floppy iris syndrome (IFIS): **billowing iris, iris prolapse, progressive miosis**. α1a blockade causes dilator atrophy, so IFIS occurs **years after stopping**; stopping pre-op does **not** help. Highest with **tamsulosin**, then alfuzosin, doxazosin and naftopidil. †If cataract surgery is awaited, defer the α1-blocker.

Drug	Vascular	Ejaculatory	IFIS
Terazosin	High	Low	Reported
Doxazosin	High (GITS lower)	Low	Lower
Alfuzosin	Intermediate	Lowest uroselective	Intermediate
Tamsulosin	Low	Dose-dependent	**Highest**
Silodosin	Lowest	**Highest**	Not well studied

KEY POINT

One efficacy, three toxicities: **α1b blockade gives vascular effects** (terazosin, doxazosin); **α1a blockade affects ejaculation and causes IFIS** (silodosin, tamsulosin); **α1d blockade gives storage benefit** (naftopidil).

11 · 5α-Reductase inhibitors

Why this matters: the only drugs that prevent retention and surgery.

- P2 Dutasteride. — *type 1+2, DHT ↓90%, 0.5 mg, PV ↓25%, AUR ↓57%/surgery ↓48%, PSA doubling, REDUCE, teratogenic*

Not in the paper answer:

- They relieve the **static component** and spare the hypothalamic–pituitary–gonadal axis, so they are the only androgen class in use (castration, antiandrogens, gonadotropin-releasing hormone analogues and aromatase inhibitors are ineffective or toxic).
- **Finasteride 5 mg** (type 2) lowers dihydrotestosterone (DHT) by †~70% and PV by 18–20%; PLESS: AUR fell 57%, surgery 55%. Head to head with dutasteride: no difference.
- **Slow**: maximum shrinkage at **6 months**; benefit needs ≥6 months.
- **Volume dependent**: best at **>40 mL** (PLESS PV >55 mL: 70% reduction). Indication: moderate–severe LUTS with PV >40 mL (EAU) or >30 mL (AUA).

- The **only class altering natural history** (fewer AUR and surgery). PSA halves, and 5-ARIs **improve PSA accuracy** for PCa.
- PCPT and REDUCE (23% fewer): fewer cancers but slightly more Gleason ≥8, so **not licensed for prevention** (possibly a volume artefact). Finasteride reduces refractory BPH haematuria.
- **AEs**: ED, lower libido and lower ejaculate volume (≈2× placebo; wane after 1–2 yr; low DHT reduces cavernosal nitric oxide synthase). **Depression** (Food and Drug Administration, FDA, 2011) and self-harm. Gynaecomastia is rare. **Post-finasteride syndrome**: symptoms persist after stopping.

> **KEY POINT**
> 5-ARIs treat the gland: slow, volume-dependent and modest for symptoms, but they prevent AUR and surgery, and they halve PSA (both a trap and a tool).

12 · Antimuscarinics

Why this matters: bladder drugs; watch the residual.

- They block detrusor **M2/M3** receptors, mainly on **afferent** pathways.
- **Indication**: predominant storage LUTS, **PVR <200 mL** and no prior AUR.
- Agents: darifenacin 7.5/15 mg, fesoterodine 4/8 mg, oxybutynin (IR 5 mg 3–4×/day; ER), propiverine, solifenacin 5/10 mg, tolterodine ER 4 mg and trospium.
- Monotherapy relieves storage without changing total IPSS or Qmax; favoured by a small prostate, PSA <1.3 and higher Qmax. In BOO, PVR rises slightly and AUR equals placebo (trials excluded PVR ≥200).
- AEs: **dry mouth**, constipation and cognitive impairment. **Oxybutynin** has most brain penetration, so avoid it in the elderly; trospium least†.

> **KEY POINT**
> Antimuscarinics treat the bladder, so every recommendation is anchored to a PVR ceiling (200 mL).

13 · β3-Agonists

Why this matters: storage relief without anticholinergic effects.

- β3 stimulation raises cyclic AMP and lowers calcium, giving **storage relaxation without weakening voiding** (Qmax unchanged).
- **Mirabegron** starts at **25 mg** and rises to **50 mg**. In BOO, PVR rose only at 100 mg; less dry mouth.
- AEs: nasopharyngitis, headache, blood pressure rise ≤1 mm Hg. **Contraindicated** in uncontrolled hypertension **≥180/110 mm Hg**; QT prolongation at 200 mg; moderate CYP2D6 inhibitor†.
- **Vibegron** 75 mg: half-life 25–38 h, no CYP2D6 inhibition; data mainly in women (EMPOWUR).

14 · PDE5 inhibitors

Why this matters: treats LUTS and erectile dysfunction together.

- Via nitric oxide and cyclic GMP they raise perfusion, relax smooth muscle, inhibit proliferation, **inhibit afferents** and reduce inflammation.

- **Tadalafil 5 mg daily** is the only licensed agent (2.5 mg is ineffective). It lowers IPSS by ≈6 (placebo 3.6), **independent of ED**, and reduces nocturia. **Qmax is unchanged**.
- First choice for LUTS with ED (AUA). AEs: headache, back pain, dyspepsia, flushing. †Contraindicated with nitrates and riociguat.

15 · Combination therapy

Why this matters: best for large glands with higher PSA.

- Short trials (6–12 months, glands <40 mL) showed no gain.
- **MTOPS** (Medical Therapy of Prostatic Symptoms): progression = AUASI rise **≥4**, AUR, UTI or urosepsis, incontinence, or creatinine rise ≥1.5 mg/dL. Placebo 17%. Number needed to treat (NNT) for combination 8.4 (≈5 if PSA >4 and PV >40).
- **CombAT** (PV ≥30, PSA ≥1.5): AUR or surgery fell two-thirds versus tamsulosin; not significant (NS) versus dutasteride.
- **Indication**: high risk, PV **>30–40 mL** and PSA **>1.5 ng/mL**. **SMART**: stop the α-blocker after 6 months.

MTOPS reduction vs placebo	Doxazosin	Finasteride	Combination
Overall progression	39%	34%	66%
≥4-point symptom rise	45%	30%	64%
AUR	NS	68%	81%
Invasive therapy	NS	64%	67%

KEY POINT

α-blockers prevent symptomatic progression only; 5-ARIs prevent AUR and surgery. The α-blocker gives fast dynamic relief; the 5-ARI gives slow static shrinkage.

Combination	Indication	Caution
α1-blocker + antimuscarinic	Persistent storage LUTS (tolterodine 4-arm; NEPTUNE)	PVR <150–200 mL only; AUR <1–3%
α1-blocker + β3-agonist	Persistent storage (MATCH)	PVR rise; limited data
α1-blocker + PDE5I	LUTS with or without ED; small gain	Hypotension; not recommended; start lowest dose
PDE5I + 5-ARI	Large prostate with or without ED	Single trial; FDA capsule 2021

16 · Phytotherapy

Why this matters: evidence is product-specific.

- **Every extract is unique** (product-specific results). Serenoa does not lower PSA.
- **Serenoa repens**: large trials of ethanolic and CO_2 extracts showed no benefit. The **hexane extract (Permixon) 320 mg daily with food** reduced nocturnal voids and raised Qmax; an EAU option to avoid drug AEs. Rare warfarin interaction.
- Pygeum is modest; others unreliable. AUA/EAU: **no general recommendation**.

17 · Follow-up

Why this matters: timed reviews catch failure early.

- Progression risk rises with PSA, PV, PVR, age and bother; dynamic AUR predictors: worsening symptoms, rising PVR. CONDUCT: bother predicts benefit.

ALGORITHM

1. Review at **4–12 weeks** (α-blocker response, AEs).
2. Review at **6 months** (5-ARI).
3. Then **annually**: score, flow, PVR; PSA on a 5-ARI if life expectancy >10 yr. WW: yearly.

18 · Acute urinary retention

Why this matters: most men avoid emergency surgery.

- P2 Catheter free trial — *spontaneous vs precipitated AUR, α-blocker before TWOC (ALFAUR), predictors of failure, recurrence, 5-ARI prevention*

Not in the paper answer:

- AUR is spontaneous (~two-thirds; worse TWOC outcome) or precipitated. Incidence is 2.2–8.5/1000 person-years.
- CombAT: AUR fell 68% versus tamsulosin but only 18% (NS) versus dutasteride; α-blockers do not prevent AUR (ALTESS, MTOPS).

ALGORITHM

1. Catheterise (†suprapubic if urethral fails).
2. Start an **α1-blocker**.
3. Do a **TWOC at ~3–5 days**.
4. If he voids, continue (add a 5-ARI for a large gland).
5. If he fails, try up to **3 TWOCs**.
6. If AUR recurs or TWOC fails: PFS and surgery *See Ch 151*.

- Emergency surgery is more morbid, so TWOC comes first; success ≈ **60%**; recurrence mostly within **6 months**.
- Failure predictors: age, severe LUTS, drained volume **>1 L**, spontaneous AUR and high-grade IPP.
- Prevention: 5-ARIs (PLESS, dutasteride, MTOPS) prevent AUR.

KEY POINT

The α-blocker rescues the TWOC (dynamic component); **the 5-ARI prevents the next episode** (static component).

19 · Landmark trials

Why this matters: these justify the drug choices above.

Trial	Design	Result	Take-home
VA transurethral resection (TURP) vs WW	Randomised trial, moderate LUTS	TURP better; most WW did not fail; crossover by bother	WW safe
PLESS	Finasteride vs placebo, 4 yr, PV ~55 mL	AUR 7% to 3%, surgery 10% to 5%; combined fall 51%	5-ARI prevents AUR/surgery

Trial	Design	Result	Take-home
Dutasteride phase III	2 yr, PV ≥30 mL	DHT fell 90%, PV 26%, AUR 57%, surgery 48%	≈ finasteride
MTOPS	Doxazosin/finasteride/combination/placebo, ≥4 yr	Progression fell 39/34/66%; AUR/surgery only with 5-ARI	Combination for high risk
CombAT	Dutasteride + tamsulosin, PV ≥30, PSA ≥1.5, 4 yr	AUR/surgery fell 66% vs tamsulosin	Combination for large gland
SMART	α-blocker withdrawal	No deterioration after 6 months	Stop α-blocker at 6 months
PCPT / REDUCE	5-ARI prevention	Fewer cancers, more high grade	Not licensed
ALFAUR	Alfuzosin vs placebo before TWOC	Success 62% vs 48%	α-blocker before TWOC

151 Minimally Invasive and Endoscopic Management of Benign Prostatic Hyperplasia

1 · Scope and principles

Why this matters: benign prostatic hyperplasia (BPH) has many operations.

- **American Urological Association (AUA) 2023 operations**: transurethral incision (TUIP), resection (TURP) and vaporisation (TUVP) of the prostate; photoselective vaporisation (PVP); prostatic urethral lift (PUL; UroLift); water vapour thermal therapy (WVTT; Rezūm); temporary implanted prostatic device (TIPD; iTind); robotic waterjet treatment (RWT; Aquablation); holmium or thulium laser enucleation (HoLEP/ThuLEP); simple prostatectomy (open, laparoscopic, robotic); prostate artery embolisation (PAE).
- Transurethral microwave therapy (TUMT) was removed in 2023 (legacy). The sham effect is large: sham endoscopy improves the International Prostate Symptom Score (IPSS) and maximum flow (Qmax), so new devices need sham-controlled trials.

KEY POINT

Volume predicts progression (acute urinary retention (AUR), surgery) but correlates poorly with symptoms. **Symptoms decide whether to operate; size and shape decide which operation.** A gland >30 cc triples AUR risk (Olmsted County).

2 · Work-up and indications for surgery

Why this matters: exclude cancer and size the gland first.

- History, examination, IPSS and urinalysis at baseline (Statement (St) 1); repeat IPSS after treatment (St 3). Bands: **0–7 mild, 8–19 moderate, 20–35 severe**.
- Before surgery, post-void residual (PVR) is **required** (St 6). **Consider** size and shape by transrectal ultrasound (TRUS), abdominal ultrasound, cystoscopy, MRI or CT (St 5), uroflowmetry (St 7), and pressure–flow study if the diagnosis is uncertain (St 8). Cystoscopy shows median lobe, stricture or tumour but is unreliable for size.
- A raised prostate-specific antigen (PSA) needs biopsy **before** BPH surgery. Counsel on failure and retreatment (St 9).

AUA Statement 26 (surgery recommended)	† European Association of Urology (EAU): absolute vs relative
Renal insufficiency from BPH Refractory retention Recurrent urinary tract infections (UTIs) Recurrent bladder stones or gross haematuria from BPH Lower urinary tract symptoms (LUTS) refractory to, or unwilling to take, medical therapy	**Absolute** (complicated BPH): refractory retention, upper-tract dilatation or renal impairment, recurrent UTI, bladder stones, recurrent haematuria refractory to 5α-reductase inhibitor (5-ARI) **Relative**: moderate–severe bothersome LUTS failing or declining medical therapy

- **Antibiotics (AUA 2019)**: one dose suffices for most transurethral cases. Cover gram-negatives and enterococci (1st/2nd-generation cephalosporin or co-trimoxazole), guided by antibiogram and urine culture. **Defer surgery if UTI is active and symptomatic**.

3 · Choosing the operation

Why this matters: each operation has a volume range.

Volume / situation	AUA 2023 options	Restrictions
<30 cc	TUIP, TURP, TUVP, PVP, HoLEP, ThuLEP, TIPD	TUIP only ≤30 cc; TIPD 25–75 cc, no obstructing middle lobe
30–80 cc	TURP, TUVP, PVP, HoLEP, ThuLEP, PUL, WVTT, RWT, TIPD	PUL: no obstructing median lobe; WVTT and RWT treat median lobes
>80–150 cc / >150 cc	HoLEP, ThuLEP, simple prostatectomy	TURP, PVP, TUVP stop at 80 cc
Higher bleeding risk	HoLEP, ThuLEP, PVP (St 43)	Otherwise hold anticoagulant/antiplatelet after liaising with prescriber
Preserve erection and ejaculation	PUL or WVTT (St 37)	Within 30–80 cc

KEY POINT

Only enucleation spans every size band; only simple prostatectomy joins it above 80 cc. TUIP alone is confined to small glands. Minimally invasive surgical therapies (MISTs) and RWT live in 30–80 cc, so volume and median lobe assessment precede every recommendation.

4 · Monopolar TURP (m-TURP): the benchmark

Why this matters: every newer operation is judged against TURP.

- P1 P2 TURP instruments and techniques; Measures to decrease complications of TURP — *resectoscope, Nesbit/floor-first resection, canoe chips, prevention of each complication*

Not in the paper answer:

- Buttocks at the table edge, so the scope reaches the anterior prostate.
- Short chips leave an irregular bed that hides bleeders.
- Over-resecting the dorsal bladder neck makes the scope go "uphill" and undermines the trigone.
- AUA: TURP for <30 and 30–80 cc (St 28), monopolar or bipolar by expertise (St 29). Monopolar irrigant is non-ionic (water, **1.5% glycine**†, sorbitol) and **hypo-osmolar**, which causes TUR syndrome.
- Keep the grounding pad away from any joint prosthesis.

ALGORITHM

1. Inspect the bladder. If tumour, resect it and defer TURP.
2. Identify orifices, neck, trigone, verumontanum (veru), sphincter; if orifices hidden, give indigo carmine or methylene blue.
3. Resect median lobe, then lateral lobes, then anterior tissue and apex last.
4. Evacuate chips, secure haemostasis, insert a large three-way catheter.

- **Anterior and apex last**: the anterior fossa has least adenoma and perforates easily, and the sphincter is tilted (anterior part more proximal).
- **Arterial bleeding** is bright and persists on filling and draining; bladder-neck bleeders show best with the bladder empty. Always fulgurate, after loop tamponade; "ricochet" bleeders may need deeper resection. **Venous bleeding** is darker and vanishes with a full bladder; once arteries are secured, use a 50–60 mL balloon with traction.

- Even routine TURP absorbs **800–1000 mL** irrigant. Catheter out at 24–48 h; stool softeners 2–4 weeks; no cycling 4–6 weeks; no sex for ≥2 weeks.
- Warn of dysuria for weeks and eschar bleeding at **1–4 weeks**. Once emptying is confirmed, treat storage symptoms with an antimuscarinic or β3-agonist; detrusor overactivity may take months. Results last about 10 years.

5 · TURP complications

Why this matters: missed TUR syndrome can kill.

- P1 P3 TUR Syndrome — *mechanism, glycine toxicity, features, risk factors, 3% saline protocol*
- Risk: gland **>45 cc**, resection **>90 min**, perforation, irrigant height. Keep the bag at about **60 cm**, because prostatic venous pressure is only about **10 mm Hg**. Na^+ **<120 mEq/L** is severe.

ALGORITHM

1. Stop or expedite surgery, secure haemostasis, stop irrigation; send Na^+.
2. If overloaded, give a †loop diuretic.
3. If severe central nervous system signs: **3% saline 100 mL** (+2–3 mEq/L), repeat ×2 at 10 min (max 300 mL).
4. Target +4–6 mEq/L; ≤12 mEq/L in 24 h (≤8–10 if presentation >48 h). Monitor in high dependency.

KEY POINT

Acute post-TURP hyponatraemia is safer to correct than chronic: the brain has not yet shed its organic osmolytes, so osmotic demyelination is less likely. Bipolar saline TURP eliminates TUR syndrome.

- **Perforation** shows glistening fat. Limited extraperitoneal: prolong the catheter. Extensive: percutaneous or open drainage. Dome: cystogram. Intraperitoneal: laparotomy.
- **Ureteric orifice**: find with a 70° lens and IV dye; with a large median lobe, resect from the midline first. **Intra-op erection**: wait; if it persists, give intracorporeal phenylephrine.

Late complication	Cause / features	Management
Bladder neck contracture (BNC)	Over-resection or over-fulguration (small glands); onset ~6 months, flow declines	Dilate, then incise (Collins knife, urethrotome or laser) over a wire into perivesical fat. Recurrent: add mitomycin C (10 mL of 0.4 mg/mL). Intractable: Y-V plasty. Obliterated: IV methylene blue and suprapubic pressure show the jet
Stricture	Meatus or fossa navicularis	Dilate meatus 2 Fr above sheath; 30 mL gel
Retrograde ejaculation	About **two-thirds**	Counsel; erection, orgasm, desire unaffected
Incontinence	Internal sphincter lost; apical over-resection causes total stress incontinence (SUI). Rare, usually transient	Veru is the distal limit; †pelvic floor exercises, then †sling or artificial sphincter
Bleeding	Early: incomplete haemostasis. Delayed (1–4 weeks): eschar	Early major: theatre. Delayed mild: fluids; †clot evacuation if retention

Late complication	Cause / features	Management
Erectile dysfunction (ED)	No conclusive causal evidence	—
Retreatment	Residual apical/anterior tissue, >80 cc, regrowth, BNC, stricture	Rises with size and pre-op retention

6 · Bipolar TURP (b-TURP)

Why this matters: saline removes TURP's main hazard.

- P2 Monopolar v/s Bipolar TURP — *saline, no TUR syndrome (NNT ≈50), less clot retention (NNT ≈20), equal efficacy to 5 y*
- Equally effective and safer, so likely to replace m-TURP as gold standard.

7 · TUIP

Why this matters: a quick, ejaculation-sparing option for small glands.

- TUIP releases capsule and bladder neck without removing adenoma. **Only <30 cc** (St 31), no median lobe (relative contraindication; EAU same). Ideal: a young man with a small gland who wants ejaculation or fertility.

ALGORITHM

1. Use a Collins knife, loop, or holmium/thulium laser.
2. Incise at 5 and/or 7 o'clock from distal to the orifice, through the bladder neck, to proximal to the veru.
3. Deepen to capsule (with or without fat); cauterise.
4. Catheter out and home on day 0–1.

- Versus TURP: shorter, less bleeding, similar symptom relief, smaller Qmax gain, **more reoperation**, less retrograde ejaculation, no BNC.

8 · Bipolar TUVP

Why this matters: good haemostasis, weaker durability.

- A broad electrode (PlasmaButton) vaporises and coagulates together, so haemostasis is excellent; no new capital, but **no specimen**. Only **bipolar** is recommended (St 32: <30 and 30–80 cc). Monopolar was abandoned: its 10 mm coagulation caused irritation, SUI and poor durability.
- Versus TURP: shorter catheter, fewer transfusions; but more dysuria, AUR and retreatment; retrograde ejaculation as common.

KEY POINT

Vaporisation trades haemostasis for a coagulated, specimen-free bed. The coagulum causes dysuria and retention; with no anatomical plane, durability is weaker. PVP makes the same trade.

9 · Laser physics

Why this matters: wavelength sets heating depth.

- P3 Lasers in BPH, including HoLEP — *wavelength/chromophore/penetration table (thulium **0.2 mm** shallowest; Nd:YAG abandoned)*

Not in the paper answer:

- Deeper penetration leaves more coagulum; hence Nd:YAG's sloughing and retention.
- Holmium is also used for TUIP.
- Diode: †940–1470 nm, †water and haemoglobin, 0.5–5 mm; needs comparative data. Safety: signage, covered windows, and eyewear for the patient too.

10 · PVP (GreenLight)

Why this matters: PVP suits anticoagulated men with glands under 80 cc.

- P4 Green light lasers — *physics, technique, AUA <80 cc + bleeding risk, GOLIATH, complications*

Not in the paper answer:

- AUA St 33: 120 W or 180 W PVP for <30 and 30–80 cc.
- Open the bladder neck first (midline or 5/7 o'clock); vaporise posterior neck level with the trigone.
- Keep working distance constant: too close damages the fibre, too far coagulates.
- Stricture, BNC and retrograde ejaculation are similar to TURP; erections minimally affected.
- No bubbles means charring, which causes dysuria. Vaporise posterior tissue in horizontal layers, never at the rectum. Avoid heavy coagulation at the neck (BNC).
- Moderate bleeding: vaporise around the feeder. Pulsatile: coagulate at ~30 W. Refractory: Bugbee (non-ionic irrigant) or resection.
- Tissue passes at 7–10 days. Dysuria lasts up to 3 months: give NSAIDs or a steroid taper, and exclude a flap or retained fibre fragment. Infection needs longer culture-directed antibiotics.
- Anticoagulated men improve equally without more transfusion.

11 · HoLEP

Why this matters: size-independent, and the only endoscopic operation that beats TURP.

- P4 P1 HoLEP — *size-independent (St 38/43), only endoscopic method superior to TURP; settings, steps, learning curve 20/50, BNC if <40 cc*

Not in the paper answer:

- Durable to 18 years: the whole transition zone (TZ) goes, whereas TURP leaves a peripheral rim.
- Morcellator injury: superficial, catheter; intraperitoneal, laparotomy and repair.
- Retained adenoma causes irritation, bleeding and infection; careless commissure splitting causes permanent SUI.

ALGORITHM

1. Incise the bladder neck at 5 and 7 o'clock (at 6 o'clock if no median lobe); carry to the veru and enucleate the median lobe.
2. Make a perimontanal inverted-U incision; open the apical plane.
3. Sweep anteriorly to 11–1 o'clock; incise the apical mucosal strip (no sphincter traction).
4. Develop the anterior plane to the neck ("Baleen" sign = bladder); release lobes into the bladder.
5. Haemostasis with a defocused beam; morcellate via nephroscope; 22 Fr three-way catheter, ~60 mL balloon.

- The smooth reflective capsule is the depth reference: too shallow leaves adenoma and feeders; too deep perforates (a few days' catheter).
- If the morcellator engages mucosa, stop cutting and suction. Retained adenoma does not dissolve, so re-morcellate.
- **Transient SUI** is common early, in proportion to volume removed; early apical release reduces it.
- Retrograde ejaculation in about **three-quarters**; median-lobe-only HoLEP preserves it in ~90%. Erections are unchanged or improved. Retreatment is rare. Anticoagulated men need slightly longer stay and irrigation; warfarin fares worse than direct oral anticoagulants.

12 · Thulium (ThuLEP) and variants

Why this matters: the same enucleation with a different laser.

- Continuous wave may carbonise. Variants: ThuVARP (vaporesection), ThuVAP (vaporisation), ThuVEP (vapo-incision), ThuLEP (mostly blunt dissection). Steps and morcellation match HoLEP.
- Function about equals HoLEP, with less bleeding than TURP (St 38 and 43). Retrograde ejaculation ~four-fifths; erections preserved.

	HoLEP	ThuLEP / ThuVEP
Learning curve	Steep (plateau ≥50)	Claimed shorter (~30)
Evidence	Largest; 18-year follow-up	Shorter follow-up

13 · MISTs

Why this matters: MISTs spare sex but are retreated more.

Ideal MIST: office-based, single session, no general anaesthesia, spares sexual function, ends medication. **Trade-off**: retreatment within 5 years exceeds surgery, and durability beyond 5 years is uncertain.

- P4 UroLift (prostatic urethral lift) — *implant, 30–80 mL no median lobe, >1.5 cm from neck, L.I.F.T./BPH6, encrustation*

Not in the paper answer:

- AUA keeps 30–80 cc and has not adopted the US Food and Drug Administration (FDA) ≤100 cc label.
- BPH6: PUL recovery was faster, but TURP also had less retreatment.
- PUL typically uses **4–6 implants**, placed anterolaterally between dorsal veins and neurovascular bundles. Remove encrusted implants with endoscopic graspers.

- **WVTT (Rezūm)** is **convective**: radiofrequency steam is injected into the TZ, and condensation releases latent heat (540 cal/g), causing 1.5–2 cm of necrosis. Capsular collagen confines the heat; total about 4 kJ.
- Give 9-s injections at 103 °C: first 1 cm distal to the neck at 3 and 9 o'clock, then every 1 cm to the veru, plus extra for a **median lobe**. Sedation or prostatic block; ~5 min. Never aim straight posteriorly near the veru; angle 45° off midline.
- **Rezūm II**: ~50% IPSS gain to 5 years; retreatment 4.4% (underestimated); **no de novo ED**; reduced ejaculate settles by 12 months. Catheter for a few days; irritation ~3 weeks. St 36/37.

KEY POINT

PUL moves tissue; WVTT ablates it. So PUL cannot treat a median lobe but needs no catheter and causes little dysuria, whereas WVTT treats median lobes at the cost of a catheter and weeks of irritation. Both preserve the bladder neck, and so antegrade ejaculation.

14 · Robotic waterjet (RWT, Aquablation)

Why this matters: TURP-level results, better ejaculation.

- A non-thermal saline waterjet ablates tissue mapped on real-time biplanar TRUS; the robot works in minutes (up to 2.4 cm deep), sparing neck, veru and sphincter. It **needs general or spinal anaesthesia, so it is not a MIST**. It suits 30–80 cc (St 39), treats median lobes, gives no specimen, and removes mainly 4–8 o'clock tissue.
- **Haemostasis is the weakness**: traction alone led to bleeding and transfusion, so **focal bladder-neck cautery** via resectoscope is now used.
- **WATER**: non-inferior to TURP to 5 years; retrograde ejaculation 10% vs 36%. Rectal injury is rare but catastrophic.

15 · True MISTs and temporary devices

Why this matters: short-term relief by design.

- True MISTs use flexible cystoscopy without sedation; durability is 12–18 months by design.
- **iTind (TIPD)**: three nitinol struts (12, 5, 7 o'clock) and a 6 o'clock anchor, left **5–7 days** to cause ischaemic remodelling, then removed via an 18 Fr catheter. St 41: 25–75 cc, no obstructing median lobe. Sham trial: responder end point met, but absolute IPSS difference not significant.
- **Optilume**: anterior commissurotomy and a paclitaxel balloon (≥5 min); single-arm data only; not on the AUA list.

16 · Prostate artery embolisation (PAE)

Why this matters: no surgery, but less effective.

ALGORITHM

1. Femoral access under local anaesthesia; pelvic angiography with the catheter balloon as landmark.
2. Pass a 1.9–2.4 Fr microcatheter into the prostatic arteries (mostly from internal pudendal; often duplicated).
3. Inject microspheres to stasis; bilateral is better.

- **Contraindications**: underactive, acontractile or neurogenic bladder; stricture or BNC; large diverticula or stones; contrast allergy; chronic kidney disease. Atherosclerosis or tortuosity may prevent completion; radiation is high.
- AUA St 40: by trained clinicians; no size limit; a role in refractory BPH haematuria. EAU: for men accepting results inferior to TURP.
- Versus TURP: inferior IPSS, Qmax and volume reduction, but shorter stay; most reintervene by 10 years.
- Complications: access haematoma or pseudoaneurysm; non-target embolisation (bladder, rectum, penis; usually conservative); post-embolisation syndrome; AUR.

17 · Overall comparison

Why this matters: this is the counselling sheet.

Procedure	Volume (AUA)	Anaesthesia / catheter	Efficacy vs TURP	Antegrade ejaculation	Key hazard	Retreatment
m-/b-TURP	<80 cc	Spinal/general (GA); 1–2 d	Benchmark	Lost ~⅔	Bleeding, TUR syndrome (mono)	Low
TUIP	<30 cc	Day 0–1	Slightly less	Mostly kept	—	Higher
b-TUVP	<80 cc	Short catheter	Similar early, less durable	Lost	Dysuria, AUR	Higher
PVP	<80 cc; anticoagulated	Day 0–1	Non-inferior (GOLIATH)	Lost ~⅔	Dysuria, UTI	Higher than TURP
HoLEP/ThuLEP	Any; anticoagulated	GA/spinal; day 0–1	**Superior** (HoLEP)	Lost ~¾–⅘	Transient SUI, morcellator	Lowest
RWT	30–80 cc	GA/spinal	Non-inferior	Mostly kept	Bleeding	Low (short data)
PUL	30–80 cc, no median lobe	Local; often none	Less	**Preserved**	Encrustation	~14% at 5 y
WVTT	30–80 cc	Sedation; few days	Less	**Preserved**	Irritation	~4–8% at 5 y
iTind	25–75 cc, no median lobe	Local; device 5–7 d	Uncertain	**Preserved**	—	Unknown
PAE	Any	Local; day case	Inferior	Largely kept	Non-target embolisation	High

KEY POINT

The "gold standard" for a given man is the operation that meets his goals safely: a shared decision on bothersome symptoms, size and shape, bleeding risk, ejaculation wishes and surgeon expertise.

18 · Landmark trials

Why this matters: evidence to quote.

Trial	Design	Result	Take-home
L.I.F.T.	PUL vs sham, 2:1, 30–80 cc	IPSS ~halved; 13.6% surgical retreatment at 5 y	PUL works, preserves sexual function
BPH6	PUL vs TURP, ≤60 cc	Composite favoured PUL; TURP better IPSS/Qmax/PVR	"Superior" refers only to composite
Rezūm II	WVTT vs sham, 30–80 cc incl. median lobe	IPSS −11.2 vs −4.3; durable 5 y; no de novo ED	Convective MIST incl. median lobe
WATER	RWT vs TURP, double-blind	Non-inferior; better in 50–80 cc; less retrograde ejaculation	Aquablation for 30–80 cc
GOLIATH	180 W PVP vs TURP, ≤100 cc	Non-inferior to 2 y; shorter catheter and stay	PVP alternative to TURP
UK-ROPE	PAE vs TURP, matched registry	PAE inferior for IPSS/QoL; shorter stay	Lesser-efficacy option

152 Simple Prostatectomy: Open, Laparoscopic and Robotic

1 · Definition and surgical anatomy

Why this matters: the capsular plane and 5/7 o'clock arteries explain every step.

- It enucleates only the benign prostatic hyperplasia (BPH) adenoma; gold standard for outlet obstruction with glands **>80 g**.
- The plane is the **surgical capsule (pseudocapsule)**: peripheral zone compressed by the enlarging transition zone *(see Ch 148)*.
- **Urethral arteries enter posterolaterally just distal to the bladder neck**, so a figure-of-eight suture goes at **5 and 7 o'clock**. The dorsal venous complex (DVC) drains to the internal iliac vein.
- Eponyms MCQ: **Freyer**, suprapubic transvesical; **Millin (1945)**, retropubic; first laparoscopic (Mariano) 2002; first robotic (Sotelo) 2008.

2 · Indications

Why this matters: large glands and bladder problems point here.

- Gland **>80–100 g** with bothersome lower urinary tract symptoms (LUTS) refractory to medical or endoscopic therapy.
- Retention, recurrent urinary tract infection (UTI), **bladder stones**, haematuria or obstructive nephropathy.
- Diverticulum needing surgery (relative); cannot lie in lithotomy for transurethral resection (TURP).

3 · Pre-operative evaluation

- Besides prostate-specific antigen (PSA), MRI, transrectal ultrasound (TRUS) volume, cystoscopy and urodynamics (atonic bladder after long retention?), check bleeding history and anticoagulants, prior pelvic surgery (adhesions dictate the route), **urine culture** (treat first) and coagulation.
- General anaesthesia is preferred; spinal is acceptable for open surgery only.

> **KEY POINT**
> Work-up asks three questions. Is the gland big enough (TRUS)? Is there cancer (digital rectal examination, PSA, MRI)? Will the bladder empty (urodynamics)?

4 · Open techniques

Why this matters: Freyer opens the bladder; Millin opens the capsule.

- P1 P2 Millin's prostatectomy — *Retzius, transverse capsulotomy ~2 cm distal to bladder neck, bladder not opened, apex divided under vision, 5/7 o'clock sutures, watertight closure; Millin vs Freyer table*

Not in the paper answer:

- Catheter: 30-mL balloon inflated to 50 mL so it cannot slip into the fossa.
- **Vesicocapsular** incision (vertical across neck and capsule): more leak and sphincter risk.
- **Freyer**: transverse **cystotomy 1–2 cm proximal to the bladder neck**; incise mucosa round the neck, avoiding orifices. A finger fractures the anterior commissure and enucleates to the

apex. It treats **median lobe, diverticula and stones**, but the apex is blind. Close in two layers, with or without a Malecot suprapubic tube (SPT).

- **Millin** also divides endopelvic fascia and puboprostatic ligaments and cauterises the superficial DVC; drawback: **DVC bleeding**.

> **KEY POINT**
> Every open variant: enter, find the pseudocapsular plane, divide the urethra at the apex **proximal to the verumontanum** without pulling on the sphincter, then 5/7 o'clock haemostasis with bladder neck mucosal advancement.

5 · Haemostasis

Technique	When / how
Pack fossa 5 min	First
5 and 7 o'clock figure-of-eight (2-0 chromic/polyglactin)	Every case; deep through proximal capsule
Bladder neck mucosal advancement ("trigonisation")	Haemostasis; prevents contracture
11 and 1 o'clock sutures	Anterior branches, occasionally
Walsh prophylactic control	Before Millin: ligate DVC and lateral pedicles, sparing neurovascular bundles (NVBs)
O'Conor plication	Diffuse bleeding: transverse sutures approximate the posterior capsule
Malament purse-string	Persistent bleeding: nylon purse-string round the neck, tied suprapubically over a button, partitions fossa from bladder; out day 2–3

> **ALGORITHM**
> 1. Catheter traction (50 mL balloon), with or without continuous bladder irrigation (CBI); exclude coagulopathy.
> 2. If bleeding continues, evacuate clot endoscopically and fulgurate.
> 3. If severe, reopen the fossa.

6 · Postoperative care (open)

- Day 2, if urine is clear: remove the Foley and clamp the SPT for a trial of void. High drain output: check **drain creatinine**. No heavy lifting for about 4 weeks.

7 · Laparoscopic and robotic

Why this matters: same enucleation, less morbidity.

- P3 Laparoscopic Trans vesical Prostatectomy — *steps, same function as open with less morbidity*

Not in the paper answer:

- Avoid posterolateral cautery: thermal NVB injury causes erectile dysfunction (ED).
- Mucosal advancement usually means **no CBI, drain or SPT**; home day 1, catheter out day 5–7.
- An extraperitoneal robotic route suits a hostile abdomen.
- **Multiport robotic simple prostatectomy (RASP)** is usually transperitoneal.

ALGORITHM

1. Vertical cystotomy; identify orifices.
2. Incise the bladder neck 3→9 o'clock under the median lobe; develop the **posterior plane** to the apex, then the lobes.
3. Divide the anterior commissure, see the veru, and transect the urethra **just proximal to the veru**.
4. Place 5/7 barbed sutures; do **vesicourethral mucosal advancement** (double-armed suture from 6 o'clock); close in two layers.

- **Single-port transvesical (STEP)**: a 2.7-cm extraperitoneal incision with **pneumovesicum 8 mm Hg** allows same-day discharge and catheter removal on day 2.

8 · Outcomes and comparisons

Why this matters: the route changes recovery, not function.

Comparison	Bottom line
Robotic vs open	Less bleeding, shorter catheter and stay, fewer complications; longer operation. Function **equal**
Single-port vs multiport	Shorter stay, less CBI and opioid; same complications
vs TURP	Lower retreatment, **no TUR syndrome**, treats stones and diverticula; but more bleeding, longer stay
†vs holmium laser enucleation (HoLEP)	†RCTs in large glands: equivalent durable outcomes; HoLEP bleeds less with shorter stay, but has a long learning curve and cannot treat diverticula *(see Ch 151)*

KEY POINT

Minimally invasive routes improve only perioperative morbidity; function is identical because enucleation is identical. Open transfusion is now **<5%**.

9 · Complications

Complication	Cause / rate	Management
Extravasation	Incomplete closure; capsular tear	If drain creatinine equals urine, keep drain and Foley
Ureteric orifice injury	Neck incision or sutures	Stent
Bladder neck contracture	Open **2–6%**; minimally invasive <1%	Dilation or Collings knife incision to 22 Fr
Storage LUTS	Weeks–months	Anticholinergic or β3-agonist
Stress incontinence	Rare	Bulking (mild); artificial sphincter (severe)
Erectile dysfunction	**3–5%**	—
Retrograde ejaculation	**80–90%**	Counsel pre-op
UTI, stricture, venous thromboembolism (VTE)	VTE <1%	Prophylaxis

KEY POINT

The adenoma lies inside the pseudocapsule and proximal to the veru. Removal loses the bladder neck (retrograde ejaculation) but spares sphincter and NVBs, so incontinence and ED are rare.

★ Putting it together: the BPH pathway

A · Man with LUTS

Why this matters: in lower urinary tract symptoms (LUTS), complications force surgery.

ALGORITHM

1. Assess: International Prostate Symptom Score (IPSS: 0–7 mild, 8–19 moderate, 20–35 severe), digital rectal examination, urinalysis, PSA if it changes management, voiding diary if nocturia; uroflow and post-void residual (PVR) before intervention *(see Ch 150)*.

2. If complications, operate: refractory retention, recurrent UTI, stones, prostatic haematuria, renal impairment.

3. If mild, watchful waiting. **4. If bothersome**, drugs (B); if they fail, a procedure (C).

B · Medical therapy

Why this matters: drugs come first.

Drug	Use	Notes
α1-blocker	First line; works in days	Dizziness, ejaculatory dysfunction; floppy iris (cataract surgery)
5α-reductase inhibitor (5-ARI)	Prostate >30–40 mL, long-term	Slow; less retention and surgery; lowers PSA by 40–50% (so double it)
α-blocker and 5-ARI	Large gland, progression risk	Best (MTOPS, CombAT)
Add antimuscarinic or β3-agonist	Persistent storage LUTS	Check PVR first
Tadalafil 5 mg daily	LUTS with ED	No nitrates
Desmopressin	Nocturnal polyuria	Check sodium, especially if ≥65

C · Choosing a procedure

Why this matters: size, bleeding risk and ejaculation decide.

Situation	Options
<30 mL	TUIP or TURP
30–80 mL	Bipolar TURP, PVP, HoLEP/ThuLEP, Aquablation; MISTs (urethral lift, water vapour)
>80 mL	HoLEP/ThuLEP; simple prostatectomy (stones/diverticulum) *(see Ch 152)*
Anticoagulation continued	HoLEP, ThuLEP, PVP
Preserve ejaculation	Urethral lift (no median lobe) or water vapour
Unfit	Long-term catheter or †prostatic artery embolisation

D · TURP complications

KEY POINT

TUR syndrome (glycine absorption) causes confusion, visual change and hyponatraemia. Stop, check sodium, give a diuretic if overloaded and 3% saline if severe, and correct slowly; bipolar prevents it. Late: retrograde ejaculation (~two-thirds), stricture (meatus, bulb), bladder neck contracture (small gland, over-resection; ~6 months), and incontinence (pelvic floor exercises, then sling or AUS) *(see Ch 151)*.